

Lecture - 9

$[A]_{n \times n}$

$$\nabla_A |A|$$

$$|A| = \sum_{j=1}^n (-1)^{i+j} A_{ij} \underbrace{|A_{\setminus i, \setminus j}|}_{\text{minor } m_{ij}} \quad \text{for any } j \in \{1, \dots, n\}$$

$$\frac{\partial}{\partial A_{kl}} |A| = \frac{\partial}{\partial A_{kl}} \sum_{j=1}^n (-1)^{i+j} A_{ij} \underbrace{|A_{\setminus i, \setminus j}|}$$

$$= (-1)^{k+l} |A_{\setminus k, \setminus l}|$$

$$= (\text{adj}(A))_{lk}$$

$$\nabla_A |A| = \left(\underline{\text{adj}(A)} \right)^T = \underline{|A| (A^{-1})^T}$$

$$A = [a_{ij}]$$

$$\text{Cofactor matrix } C = [C_{ij}]$$

$$C_{ij} = (-1)^{i+j} m_{ij}$$

$$\text{adj}(A) = C^T$$

$$P = A \text{adj}(A)$$

$$P_{ij} = \sum_{k=1}^n a_{ik} C_{jk}$$

① Case $\therefore i = j$

$$P_{ii} = \sum_{k=1}^n a_{ik} C_{ik}$$

$$P_{ii} = a_{i1}C_{i1} + a_{i2}C_{i2} + \dots + a_{in}C_{in}$$

$$P_{ii} = |A|$$

② Case :- $i \neq j$

$$P_{ij} = \sum_{k=1}^n a_{ik} C_{jk}$$

- It is the determinant of a matrix obtained by replacing the j th row with i th row.
- The new matrix will have 2 identical rows
- The determinant of matrix with 2 identical rows is 0.

$$P_{ij} = 0$$

$$P = A \operatorname{adj}(A) = \begin{bmatrix} |A| & 0 & 0 & \dots & 0 \\ 0 & |A| & 0 & \dots & 0 \\ 0 & 0 & |A| & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & |A| \end{bmatrix}$$

$$A \cdot \operatorname{adj}(A) = |A| \cdot I$$

$$\operatorname{adj}(A) = |A| \cdot (A^{-1})$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 4 & 5 & 6 \end{bmatrix}$$

$$\Rightarrow \text{Determinant} = 0$$

$$\neq \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\text{adj}(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A \cdot \text{adj}(A) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= \begin{bmatrix} ad - bc & -ab + ab \\ cd - cd & ad - bc \end{bmatrix}$$

$$= \begin{bmatrix} |A| & 0 \\ 0 & |A| \end{bmatrix}$$

$$= |A| \cdot I$$

$$f(A) = \log |A|$$

$$\frac{\partial \log |A|}{\partial A_{ij}} = \frac{\partial \log |A|}{\partial |A|} \cdot \frac{\partial |A|}{\partial A_{ij}}$$

$$= \frac{1}{|A|} \cdot \frac{\partial |A|}{\partial A_{ij}}$$

$$= \frac{1}{|A|} \cdot |A| (A^{-1})^T$$

$$= (A^{-1})^T = A^{-1} \begin{bmatrix} I & A \\ 0 & \text{is symmetric} \end{bmatrix}$$

Eigen values as Optimization:-

$$\begin{array}{ll} \max_{x \in \mathbb{R}^n} x^T A x & \text{s.t. } \|x\|_2^2 = 1 \end{array}$$

$\downarrow$ objective fn. $\downarrow$ Constraint

[A is symmetric]

The Lagrangian in this case:-

$$\mathcal{L}(x, \lambda) = x^T A x - \lambda (x^T x - 1)$$

$\rightarrow$ Lagrangian multipliers

$$\nabla_x \mathcal{L}(x, \lambda) = 2Ax - 2\lambda x \rightarrow 0$$

$$Ax = \lambda x.$$

The only points which can possibly maximize or minimize $x^T A x$ assuming $\|x\|_2^2 = 1$ are the eigenvectors of A.

$$f(x): \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$$

$$x \in \mathbb{R}^{m \times n}$$

$$A \in \mathbb{R}^{m \times n}$$

$$\textcircled{i} \quad f(x) = \text{trace}(A^T x) = \langle A, x \rangle$$

$$\frac{\partial f(x)}{\partial x} = A$$

$$\textcircled{i} \quad f(x) = \text{trace}(x^T A x)$$

$$\frac{\partial f(x)}{\partial x} = (A + A^T)x$$

$$\textcircled{ii} \quad f(x) = \frac{1}{2} \|Ax - B\|_F^2$$

$$\frac{\partial f}{\partial x} = A^T (Ax - B)$$

$$\|A\|_F = \sqrt{\sum_{i=1}^n \sum_{j=1}^n a_{ij}^2}$$

$$\textcircled{iv} \quad f(x) = \frac{1}{2} \text{trace}(B^T x^T x B)$$

$$\frac{\partial f}{\partial x} = x B B^T$$

$$\textcircled{v} \quad f(x) = \frac{1}{2} \text{trace}(B^T x^T A x B)$$

$$\frac{\partial f(x)}{\partial x} = \frac{1}{2} (A + A^T) x B B^T$$